Where Does Abandoned Cropland Go? A Decade of Land-Cover Trajectories in Ohio

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AAG American Association of Geographers

Online Maps & Figures

We ask: **Q1.** Where does abandoned cropland go within ten years? **Q2.** Which pathways does it follow, and where? **Q3.** Do the factors used to predict recultivation also separate these pathways?

Why trajectories?

Abandonment is a transition, but large studies see a snapshot or two futures.

- **Widespread.** About 101 Mha of cropland were abandoned worldwide in 1992–2020 (Zheng et al., 2023) and 12.3 Mha in the conterminous U.S. in 1986–2018 (Xie et al., 2024).
- **A transition, not an end state.** Fields may regrow forest, turn to grass or pasture, return to crops, or be built on (Munroe et al., 2013; Crawford et al., 2022).
- **Simplified at scale.** Large studies map one snapshot (Xie et al., 2024) or model two futures, recultivation or reforestation (Zheng et al., 2023).
- **Now traceable.** Annual 30 m land cover follows each abandoned pixel year by year (Fig 0), catching changes that start-and-end maps miss (Auch et al., 2022).

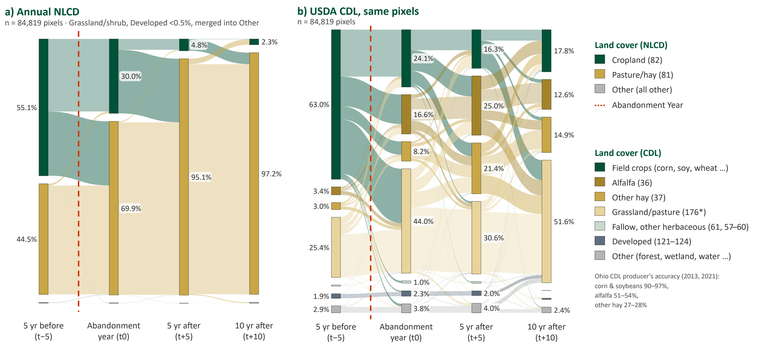


Fig 0. Reading a trajectory: pasture/hay-transition pixels abandoned in 2013 (n = 84,819), followed from t−5 to t+10. NLCD (a) shows one switch to pasture/hay; CDL (b) mixes crops, hay and grass.

Data & approach

1. **Abandoned cropland.** Xie et al. (2024), 30 m, abandoned 2001–2014 and not recultivated. Ohio: 1.86 M pixels (≈127,000 ha).
2. **× Annual NLCD 1996–2024,** grouped into cropland, pasture/hay, developed and other.
3. **Aligned to the abandonment year (t0),** from t−5 to t+10.
4. **k-means (k = 5) → five types.** Pasture/hay splits into long-term (already pasture/hay before t0) and short-term (new after t0).
5. **Models.** Each main type's share in ≈300 m cells (n = 55,784) is regressed on Zheng et al.'s factors (Table 1) with a spatial error model.

Table 1. Zheng et al.'s nine factors, rebuilt with U.S. data

Factor	U.S. measure
1 Agro-ecological suitability	Crop calorie yield (≈9 km window)
2 Market accessibility	Distance to major road (OSM)
3 Distance to urban area	Distance to Census urban area
4 Population density (4a, 4b)	Ambient (4a) and residential (4b)
5 Cropland density	Crop share, ≈300 m window (CDL)
6 Distance to stable cropland	Distance to land cropped every year (CDL)
7 Abandoned cropland density	Abandoned share, ≈300 m window
8 Disaster risk	FEMA risk score
9 Agricultural economic loss	FEMA expected agricultural loss

Q1 Where does it go?

After ten years, 85% was still farmland; only 1.5% had become forest.

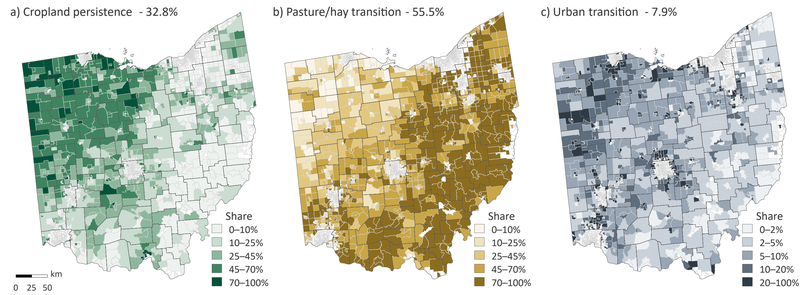


Fig 1. Share of each trajectory type by census tract (tracts with ≥50 abandoned pixels; n = 1,084). Title percentages are Ohio-wide.

- At t+10, abandoned cropland is mostly pasture/hay (61%) or still cropland (24%); 11% is developed and 4% other.
- Three main pathways: pasture/hay transition (55.5%), cropland persistence (32.8%) and urban transition (7.9%).
- Compared with the other 11 Midwest states, almost none becomes grassland or shrub (vs. 23%), and more is developed (11% vs. 4%).

Q2 Which pathways, where?

Cropland persistence dominates the northwest, long-term pasture/hay the Appalachian east and south; central Ohio is mixed.

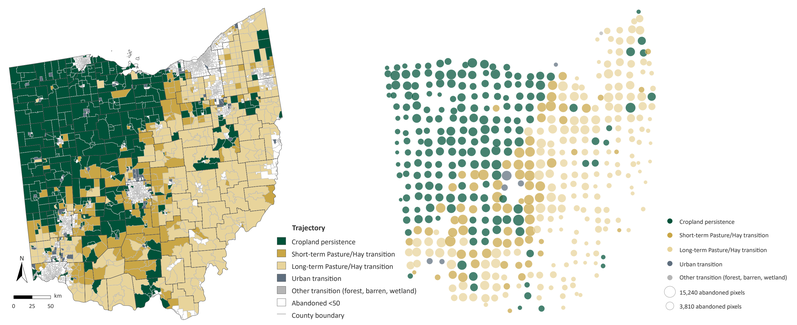


Fig 2. Most common of the five types by census tract (left; n = 1,084) and by 15 km cell (right; n = 470; circle area scales with √pixels). Northwest: west of 83.4°W and north of 40.2°N; Appalachia: 32 ARC counties.

- **Pasture/hay transition (55.5%).** Two halves: long-term pixels were pasture/hay before abandonment; short-term pixels switched within five years after it.
- **Cropland persistence (32.8%).** NLCD still maps cropland, but the USDA crop map (CDL) shows hay or grass on most of it (58% of pixels abandoned 2010–14).
- **Urban transition (7.9%).** Mostly low-intensity development; almost half was already developed five years before abandonment.
- **Where.** Cropland persistence leads in 80% of northwestern tracts, long-term pasture/hay in 69% of Appalachian tracts.

Q3 What separates the pathways?

Landscape context, not agro-ecological suitability, separates the pathways.

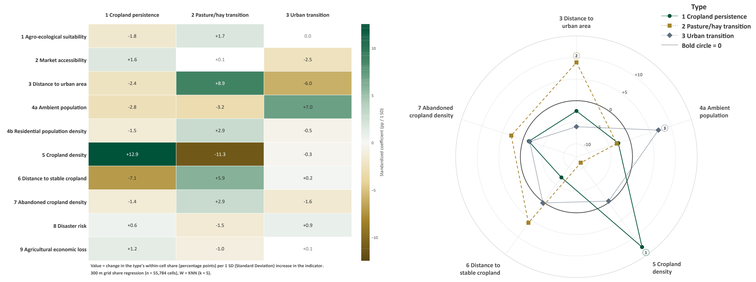


Fig 3. Change in each type's share (percentage points) per 1 SD of each indicator, from spatial error models (n = 55,784 cells). Left: all ten indicators (white = not significant). Right: the five strongest (bold circle = 0).

- **Cropland persistence** is higher where surrounding cropland is dense (+12.9) and stable cropland is near (−7.1 per SD of distance).
- **Pasture/hay transition** is higher where cropland is sparse (−11.3), far from urban areas (+8.9) and far from stable cropland (+5.9).
- **Urban transition** is higher with larger ambient population (+7.0) and near urban areas (−6.0).
- Agro-ecological suitability and the two disaster indicators matter little (all $|\beta| < 2$).

Take-home

1. **In Ohio, abandoned cropland mostly stays in farming, as hay or pasture.** Forest regrowth within a decade is rare.
2. **Pathways are regionally sorted** and follow landscape context: nearby cropland and urban areas.
3. **Trajectories reveal what snapshots and two-option scenarios miss,** including where datasets disagree.

Caveats & next steps

- **Land cover is not land use.** NLCD cannot tell idle grass from managed hay, and Xie et al. count a switch to hay as abandonment.
- **The abandonment map has errors.** CDL shows crops continuing on about 1 in 5 pixels abandoned in 2010–14.
- **Window and sample.** Recultivated fields are excluded by design, and ten years is short for forest to return.
- **Next.** Model the three shares jointly, rebuild cropland indicators without the pixel itself, screen out mapping errors, and extend to the conterminous U.S.

References

Auch et al. (2022) *Land* 11:298 · Crawford et al. (2022) *Sci. Adv.* 8:eabm8999 · Munroe et al. (2013) *Curr. Opin. Environ. Sustain.* 5:471–476 · Xie et al. (2024) *Environ. Res. Lett.* 19:044009 · Zheng et al. (2023) *Nat. Commun.* 14:6083 · USGS (2024) Annual NLCD Collection 1 Science Products, ver. 1.1, doi:10.5066/P94UXNTS · USDA NASS CDL 2008–2023